Accelerated Dynamic Renal Phase-Contrast MRI using A Score-based Diffusion Network

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Outline

1. Background
2. Methods
3. Results
4. Conclusions

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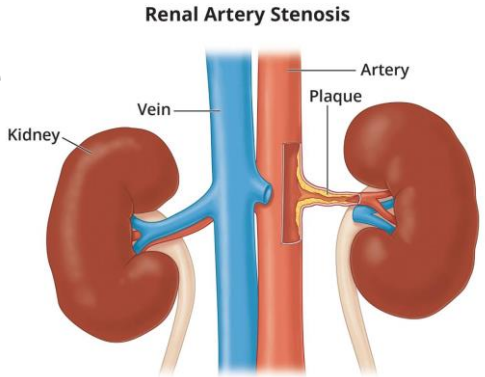
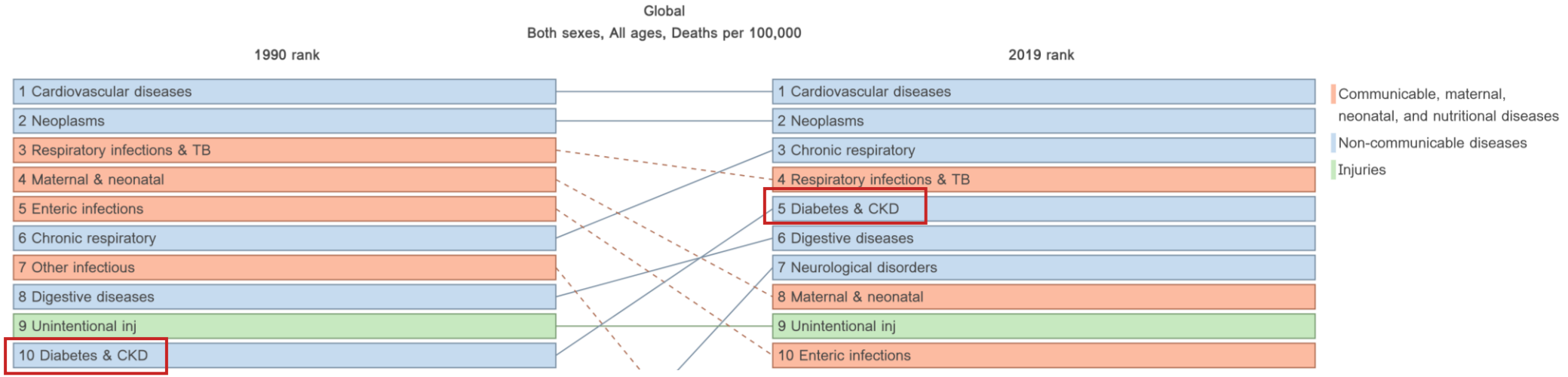
Background

- Kidney diseases are the **10th** leading cause of death worldwide.(WHO)
- Chronic kidney disease statistics worldwide from 2019 show us that over **850 million** people around the world live with some type of kidney disease. (Oxford Academic)
- The mortality rate of CKD **increased 41.4%** between 1999 and 2017. (GBD)
- Renal artery stenosis (RAS) is a frequently encountered problem in clinical practice.





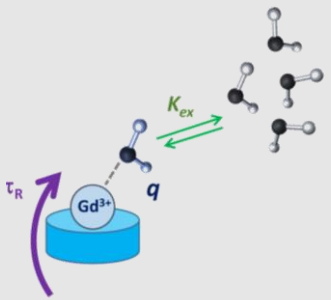
[1] Global Burden of Disease, https://vizhub.healthdata.org/gbd-compare/
[2] National Institute of Diabetes and Digestive and Kidney Diseases, https://www.niddk.nih.gov/health-information/kidney-disease/renal-artery-stenosis

Background

Clinical problems

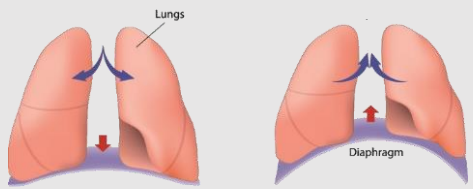
Contrast agent

- Gadolinium-based contrast agent may cause nephrogenic systemic fibrosis



motion

- Respiratory motion may cause artifacts
- It’s not easy to hold breath for too long
- The position of the breath-holding organ is different each time



Scan time

- Long scan time increases patient discomfort

[1] Lacerda S. Inorganics. 2018; 6(4):129.
[2] https://portmelbournephysio.com.au/diaphragmatic-breathing/

Background

Overall Aim

- Free-breathing renal artery blood flow accelerated imaging based on Phase-Contrast MRI

Specific Objectives

- Data acquisition for dynamic PC-MRI
- Accelerated imaging algorithm

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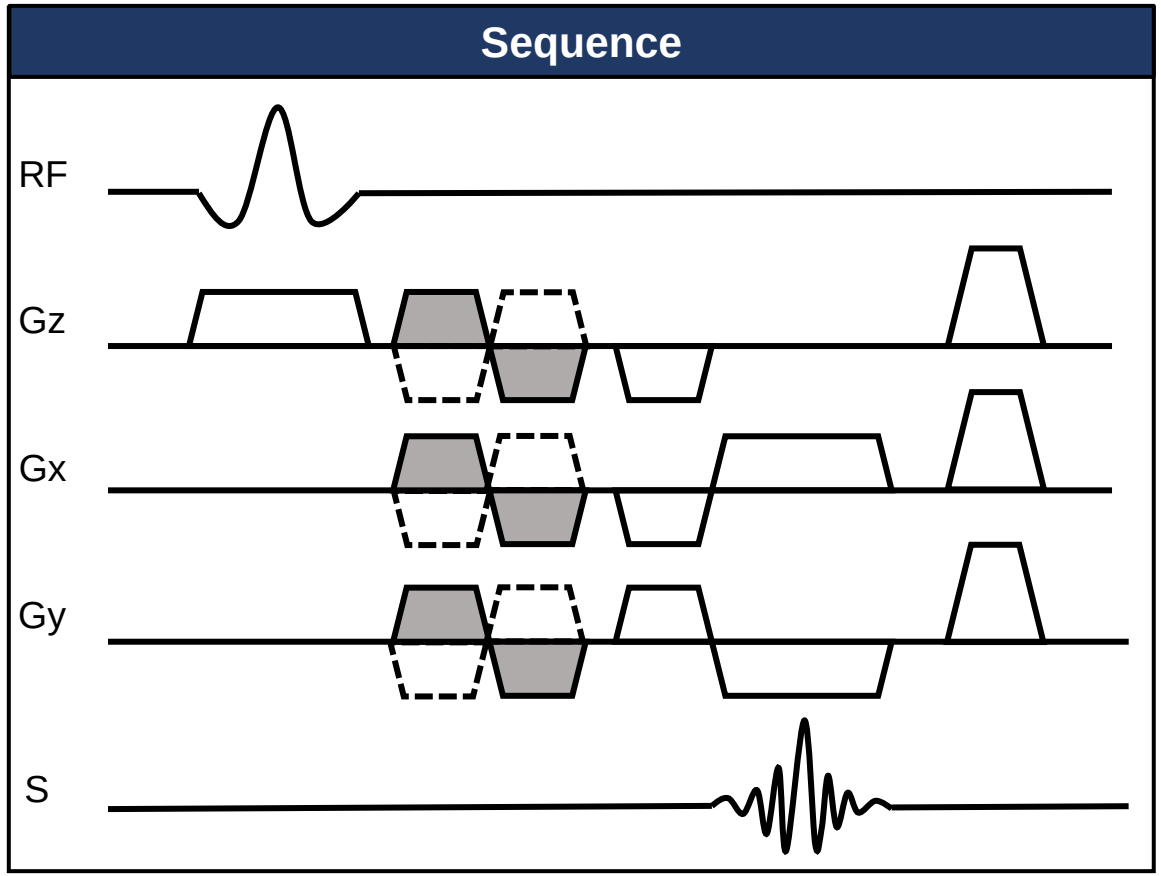
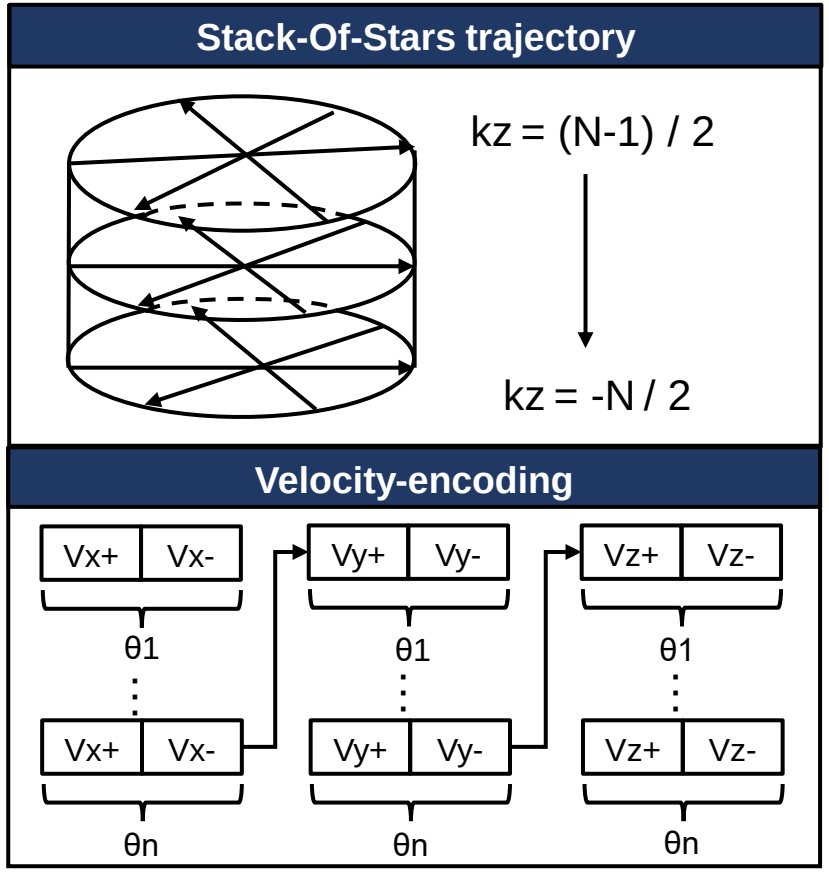
Methods

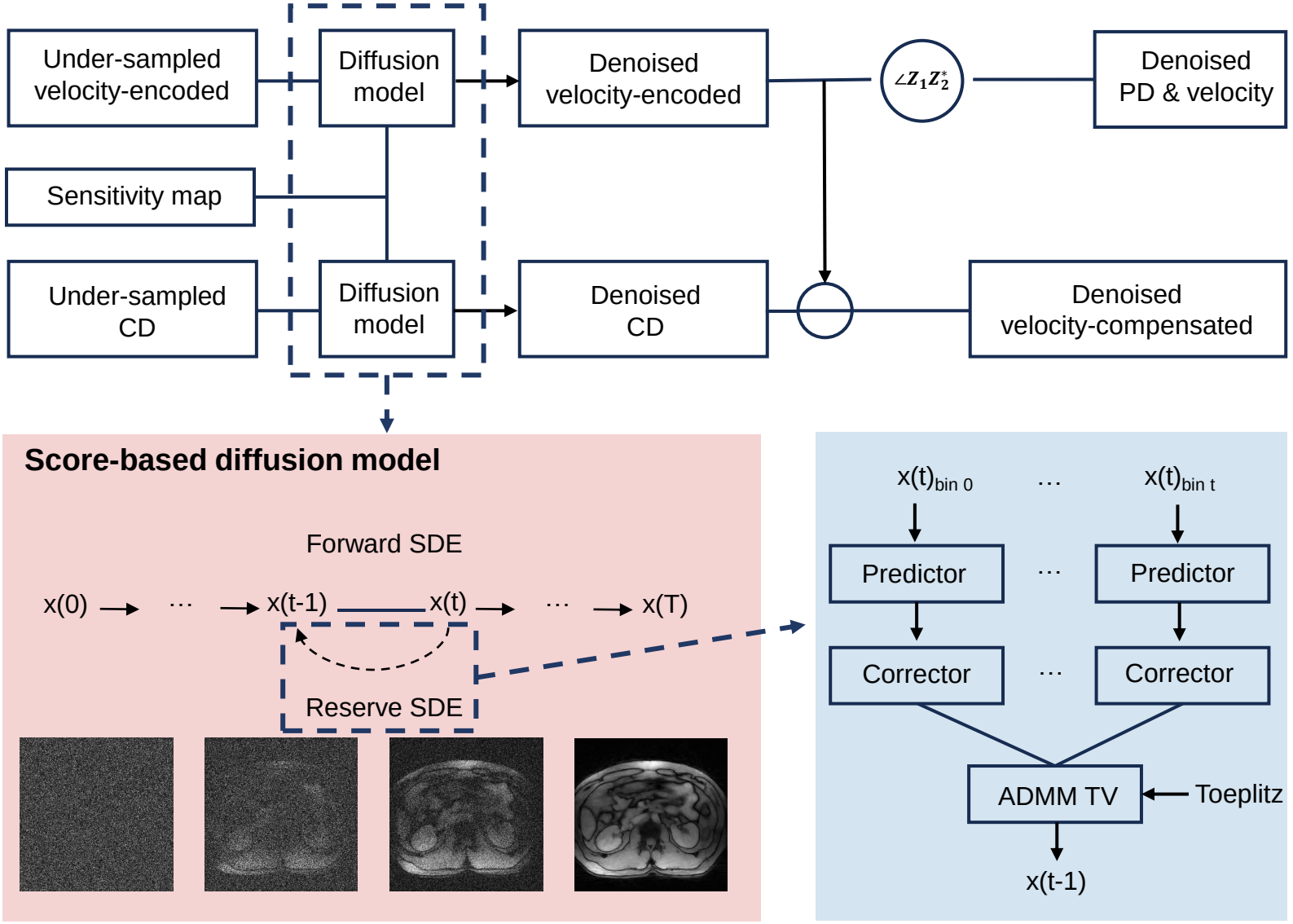




- Golden-angle Stack-Of-Stars (SOS) radial GRE sequence
- Same angle for each pair of bipolar in two consecutive interleaves
- Self-navigation

Methods

Framework



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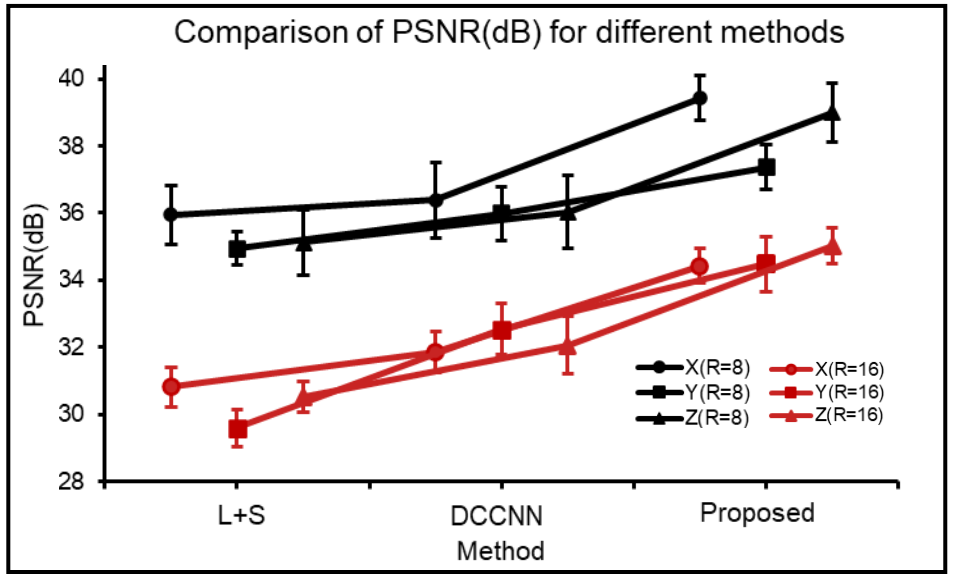
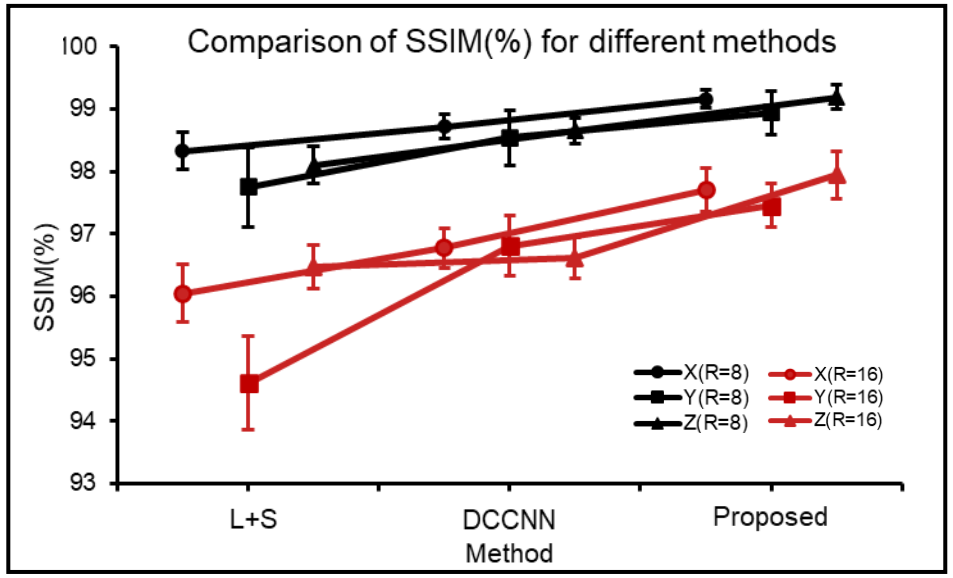
Results

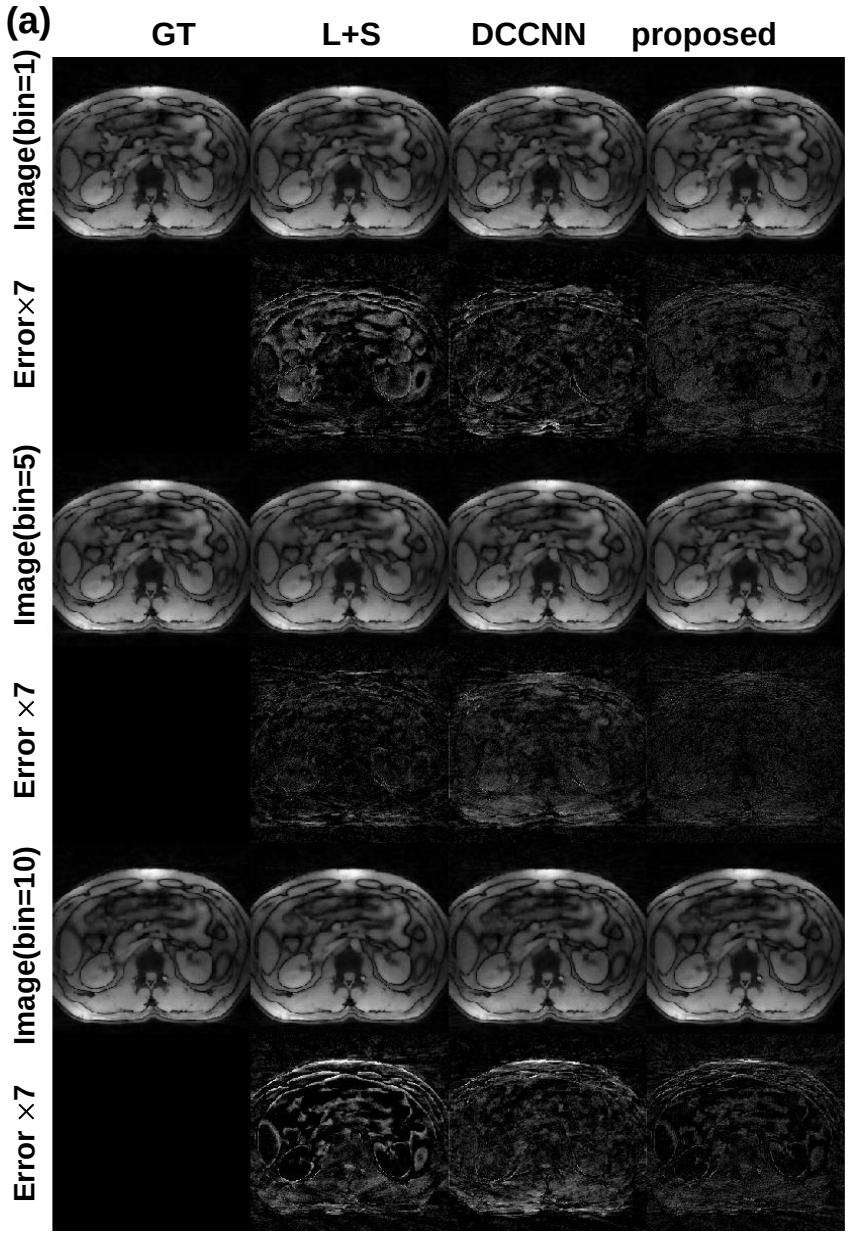
Table: PSNR and SSIM of velocity-encoded image for L+S, DCCNN and proposed method

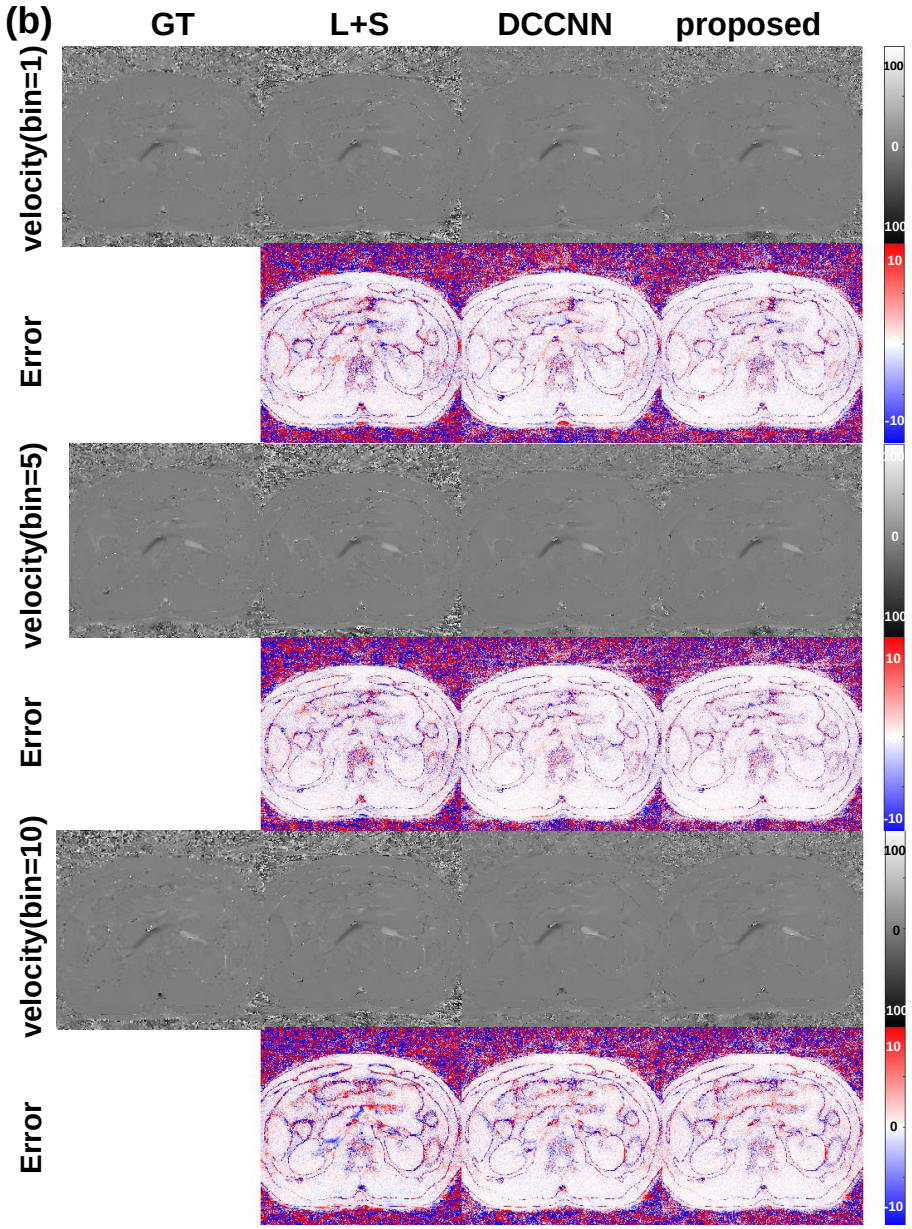
		R=8		R=16	
		PSNR	**SSIM**	**PSNR**	**SSIM**
L+S	X	35.94 ±0.89	98.33 ±0.29	30.81 ±0.58	96.05 ±0.46
	Y	34.94 ±0.97	97.75 ±0.64	29.60 ±0.55	94.61 ±0.75
	Z	35.13 ±0.49	98.10 ±0.30	30.52 ±0.47	96.47 ±0.35
DCCNN	X	36.38 ±1.12	98.72 ±0.20	31.85 ±0.60	96.77 ±0.32
	Y	35.98 ±1.10	98.54 ±0.44	32.53 ±0.76	96.81 ±0.48
	Z	36.03 ±0.81	98.65 ±0.21	32.05 ±0.86	96.62 ±0.33
Proposed	X	**39.43 ±0.67**	**99.16 ±0.15**	**34.43 ±0.52**	**97.71 ±0.35**
	Y	**37.37 ±0.88**	**98.94 ±0.35**	**34.47 ±0.81**	**97.45 ±0.60**
	Z	**39.00 ±0.65**	**99.19 ±0.20**	**35.03 ±0.54**	**97.95 ±0.38**





Results



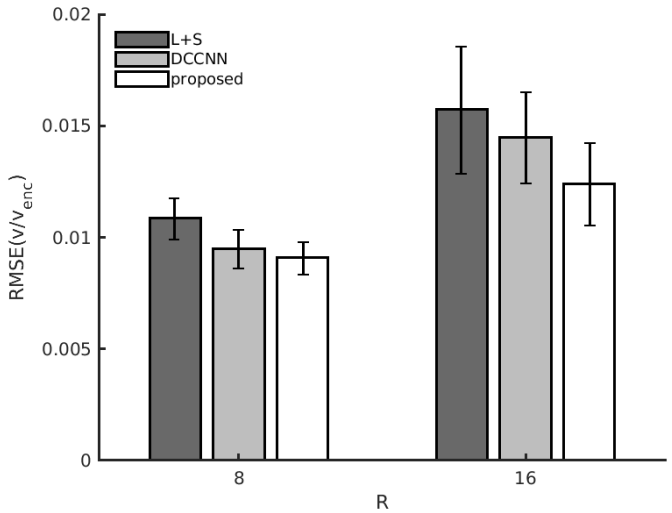


- Comparison of results between different methods at R=16, Slice 10 and Bin 1, 5, 10, including reconstructed images and their errors in comparison with the ground truth of (a) velocity-encoded images and (b) velocity maps.

Results

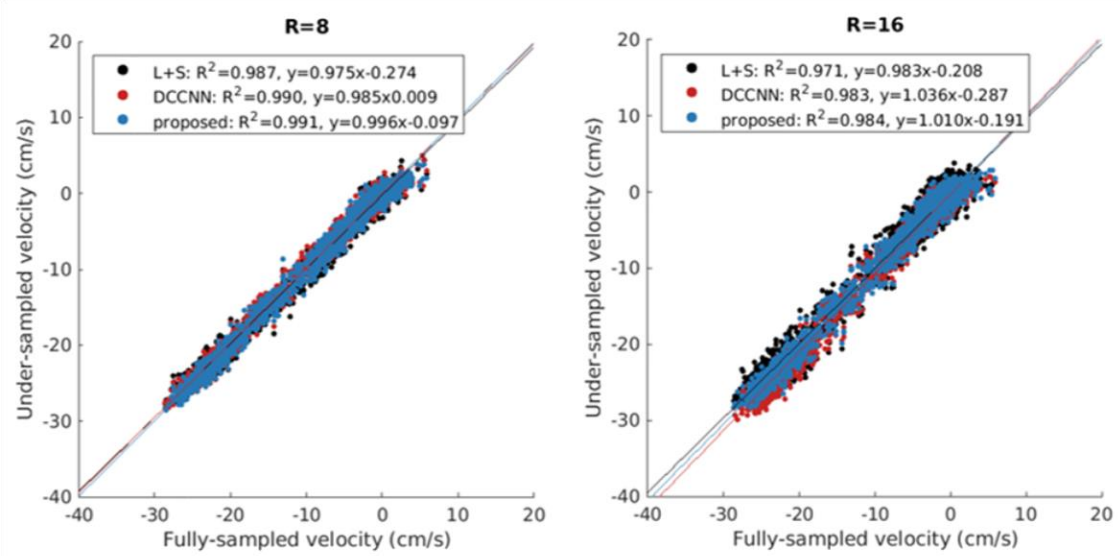
Root Mean Square Error

	L+S	DCCNN	Proposed
R=8	0.0108±0.0009	0.0095±0.0009	**0.0091±0.0007**
R=16	0.0157±0.0029	0.0145±0.0020	**0.0124±0.0019**

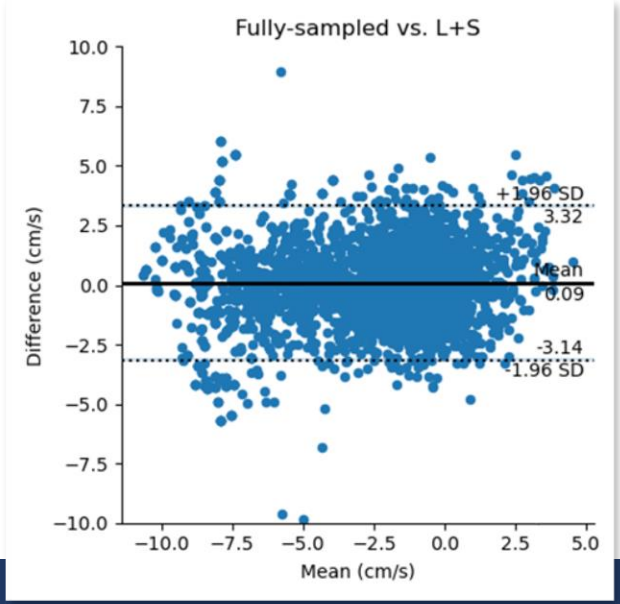


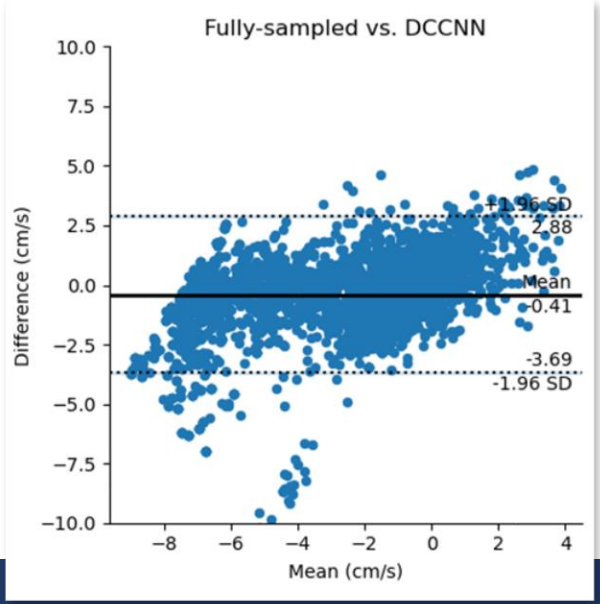
Linear regression

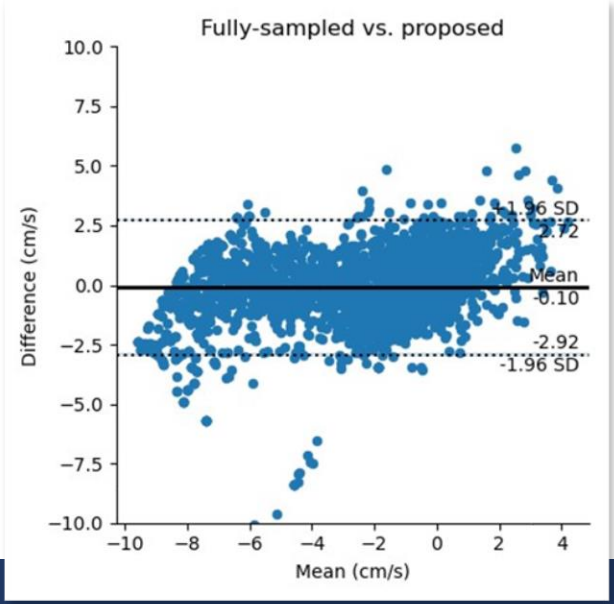
- Both slope and R^2 Shows a strong linear relationship with the ground truth in ROI along x



Results







Bland-Altman plot

	L+S	DCCNN	Proposed
Mean	**0.09**	0.41	0.10
95% CI	[-3.14, 3.32]	[-3.69, 2.88]	**[-2.92, 2.72]**

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Conclusions

- Incorporate temporal constraints for score-based diffusion model to solve dynamic PC-MRI reconstruction
- The ability to significantly reduce scan time
- Train network with non-PC-MRI data
- Potential for unsupervised learning and prospective studies

Thank you

If you have any questions, please contact us by sending emails to

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